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MODEL SELECTION AND VALIDATION REPORT ( general_kde_feve model )
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## OBSERVATIONS:3451

## SANITY: OK - all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

## MODEL SUMMARY ( general_kde_feve model )

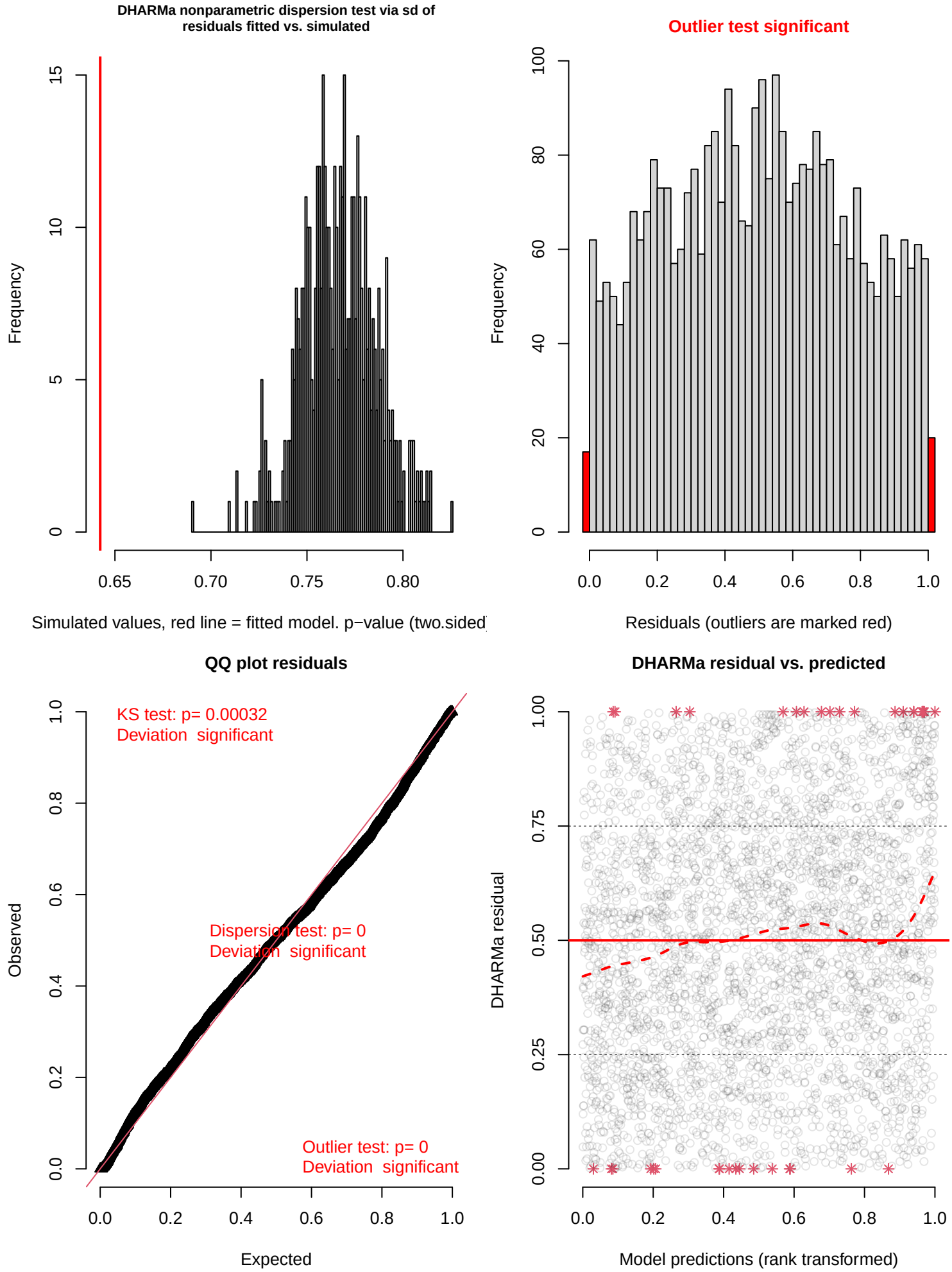
Spatiotemporal model fit by REML ['sdmTMB']
Formula: fixed_formula
Time column: year
Family: gaussian(link = 'identity')

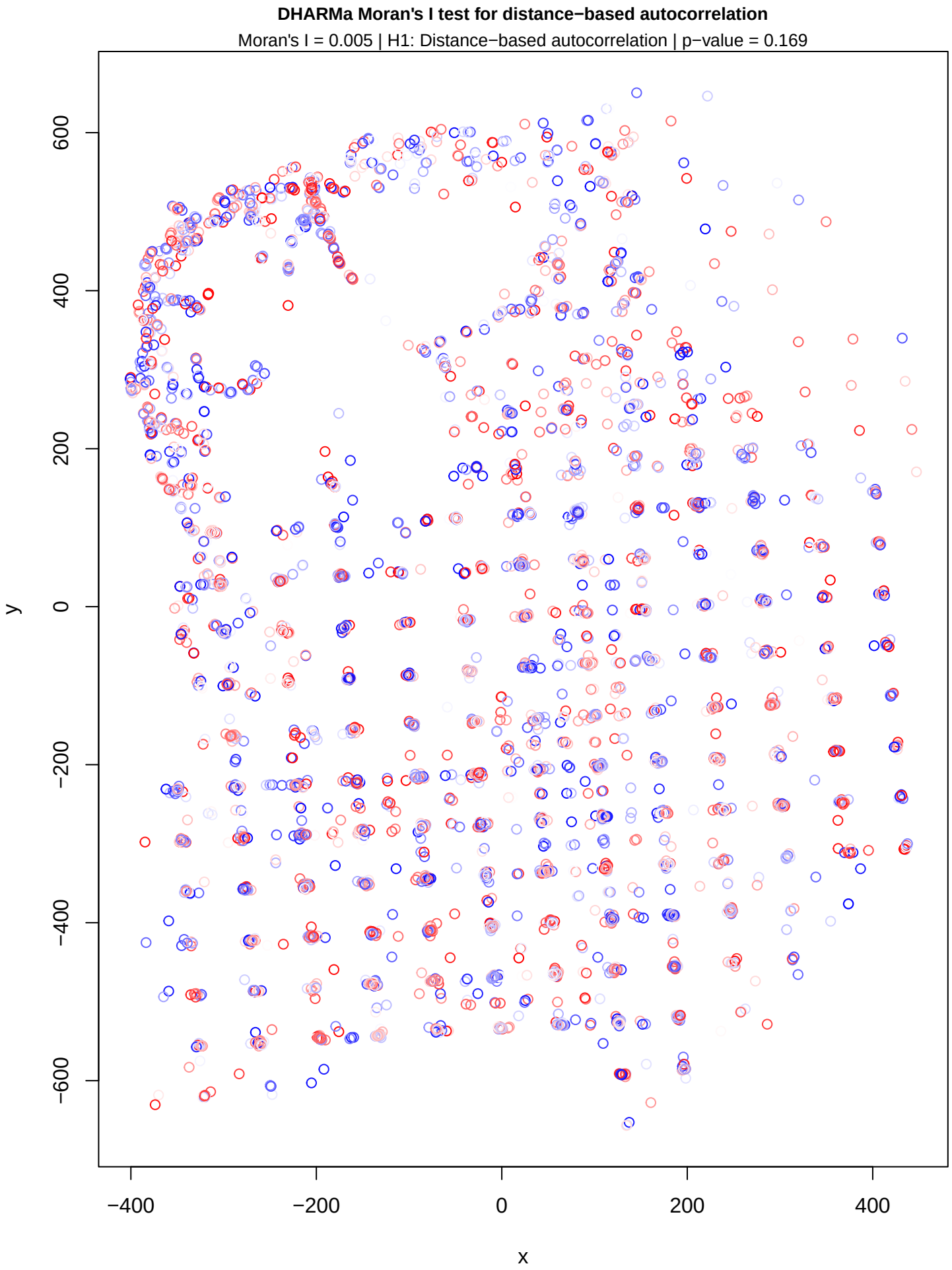
Conditional model:
              coef.est coef.se
(Intercept)      -0.06    0.13
sbt_mean          0.02    0.05
sbt_sd           -0.08    0.03
sst_sd            0.02    0.03
sic_mean         -0.09    0.05
log_chla_mean    -0.01    0.02
depth            -0.13    0.03

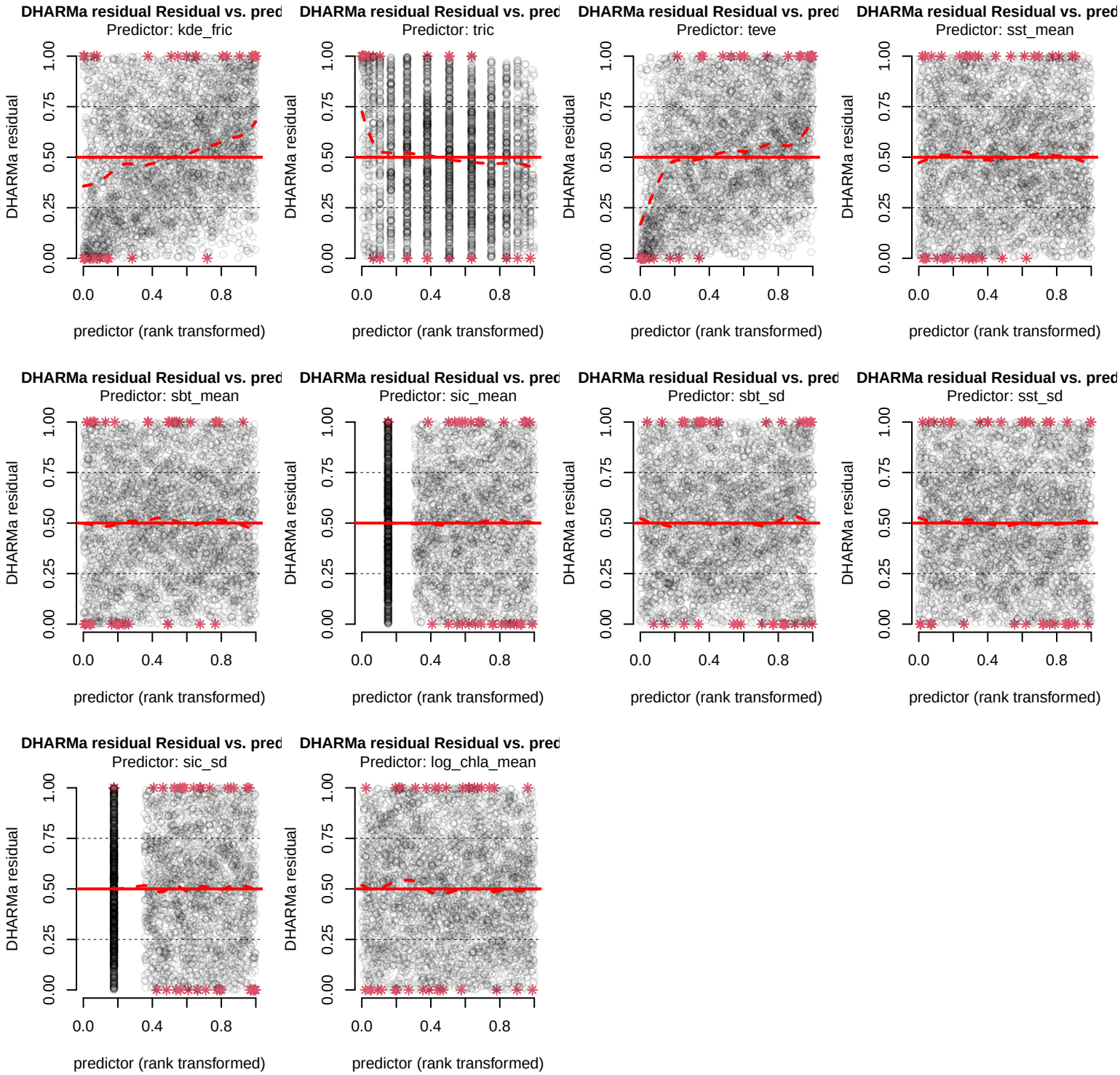
Dispersion parameter: 0.85
Spatiotemporal AR1 correlation (rho): 0.85
Matérn range: 326.82
Spatiotemporal marginal AR1 SD: 0.57
REML criterion at convergence: 4573.928

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS
                                                   df      AIC  delta_AIC
spatiotemporal_random_fields_ar1                   11 9169.856   2.710284
spatiotemporal_random_fields_ar1_with_anisotropy 13 9172.567 135.999339
spatiotemporal_random_fields_iid_with_anisotropy 12 9308.566   6.354121
spatiotemporal_random_fields_iid                   10 9314.920   9.285722
spatial_random_fields                              10 9324.206   3.734602
spatial_random_fields_with_anisotropy              12 9327.941 287.161341
error_only                                          8 9615.102         NA
```


DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
5
10
15
0.65
0.70
0.75
0.80
Simulated values, red line = fitted model. p−value (two.sided
Outlier test significant
Frequency
0
20
40
60
80
100
0.0
0.2
0.4
0.6
0.8
1.0
Residuals (outliers are marked red)
QQ plot residuals
KS test: p= 0.00032
Deviation significant
Dispersion test: p= 0
Deviation significant
Outlier test: p= 0
Deviation significant
Observed
0.0
0.2
0.4
0.6
0.8
1.0
Expected
DHARMa residual vs. predicted
DHARMa residual
0.00
0.25
0.50
0.75
1.00
Model predictions (rank transformed)


DHARMa Moran's I test for distance−based autocorrelation
Moran's I = 0.005 | H1: Distance−based autocorrelation | p−value = 0.169
y
600
400
200
0
−200
−400
−600
x
−400
−200
0
200
400


DHARMa residual Residual vs. pred
Predictor: kde_fric
Predictor: tric
Predictor: teve
Predictor: sst_mean
Predictor: sbt_mean
Predictor: sic_mean
Predictor: sbt_sd
Predictor: sst_sd
Predictor: sic_sd
Predictor: log_chla_mean
DHARMa residual
predictor (rank transformed)
0.00
0.25
0.50
0.75
1.00
0.0
0.4
0.8